

Energy Monitor — Portable Desktop Edition

Data Center Energy & Facility Monitor as a fully offline, portable Windows application. The Excel workbook `RST_Dashboard.xlsx` is the primary database; every dashboard, forecast, and report reads from it, and every save writes back into it — with an automatic timestamped backup before each write.

Internal enterprise tool. No installer, no license system, no online updates, no telemetry.

Quick start

- Extract `EnergyMonitor_Portable.zip` anywhere (Desktop, `D:\Tools`, a USB drive — any folder you can write to).
- Double-click `Energy Monitor.exe`.
- That's it. The app finds `RST_Dashboard.xlsx` beside the executable and opens it automatically. No installation, no administrator rights, no internet connection required.

Portable folder layout

```
Energy Monitor/
├─ Energy Monitor.exe      ← the application
├─ RST_Dashboard.xlsx      ← your database (macros, pivots & charts intact)
├─ config/                 ← config.json + crash-recovery journal
├─ backup/                 ← automatic timestamped backups (rotated)
├─ logs/                   ← app.log (size-rotated)
├─ exports/                ← default folder for JSON/CSV exports
└─ docs/                   ← this documentation
```

Everything stays inside this folder. The app never writes to AppData, the registry, or browser Local Storage.

What it does

Area	Features
Data entry	Monthly UPS loads, air-conditioning energy, DC panels, building energy & cost — with historical-edit confirmation
Reports	Executive Dashboard, Dashboard Summary, Benchmark, Forecast, Historical Explorer, Trend analytics, Smart Insight
Workbook	Open / Save / Save As / Recent files / drag & drop / "Open With"
Safety	Backup before every save, atomic writes, lock detection (file open in Excel), crash recovery, restore-from-backup
Integrity Center	Duplicates, missing months, missing devices, invalid IDs, blank rows, workbook health, last-validation time
Settings	Default workbook, startup behavior, auto-save interval, backup retention, theme (dark/light), language (TH/EN), optional Google Sheets sync

Your workbook stays **your** workbook

Saves are performed by surgical patching of only the four log sheets.

Everything else in `RST_Dashboard.xlsx` — **VBA macros, pivot tables, charts, the dashboard sheet, cell styles, the "4th Floor Electricity Cost" column and the calculated rate column** — is preserved byte-for-byte. Excel Tables are extended so the dashboard formulas and pivot caches pick up new months, and the workbook is flagged to recalculate and refresh pivots on next open in Excel.

If the workbook is open in Excel while you save, the app detects the lock and offers **Retry / Save As / Cancel** — your edits are journaled and can never be lost, even across a crash.

Documentation

- [User Guide](#) — day-to-day usage
- [Developer Guide](#) — code architecture, building, tests
- [Deployment Guide](#) — building and rolling out the ZIP
- [Architecture](#) — diagrams: system, IPC, data flow
- [Migration Summary](#) · [Regression Report](#) · [Production Readiness](#)